

Dataset: Adult Income (UCI)
Primary Evaluation Metric: Precision
Final Model: Histogram Gradient Boosting (Tuned + Isotonic Calibration)

1. Objective

This report summarizes the tuning, diagnostics, calibration and final evaluation performed during Week 4 Day 4. The objective was to transform the best-performing Day 3 model into a reproducible deployment-ready pipeline while optimising the project's primary metric (Precision).

2. Hyperparameter Search Summary

RandomizedSearchCV (5-fold Stratified Cross Validation) was performed for both candidate models.

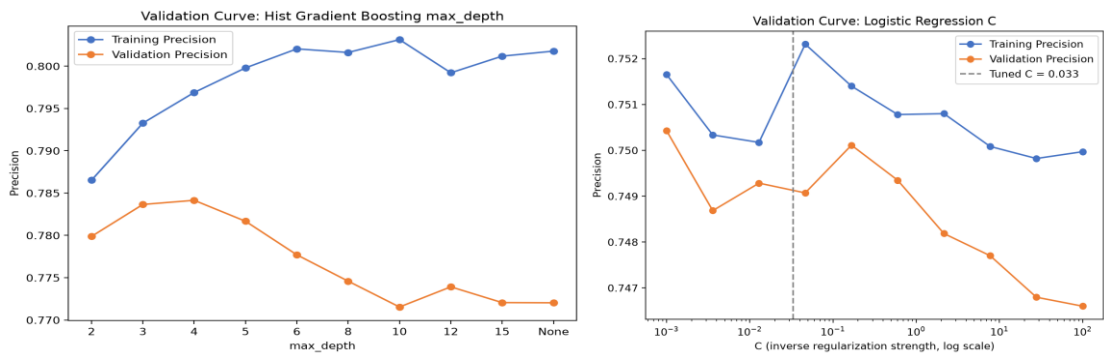
Model	Best CV Precision	Selected
Logistic Regression	0.7537	No
Histogram Gradient Boosting	0.8257	Yes

Best Histogram Gradient Boosting Hyperparameters:

Hyperparameter	Value
learning_rate	0.0129
max_depth	9
max_iter	108
l2_regularization	1.0256

3. Model Diagnostics

Learning curves indicated mild variance and good generalisation for Histogram Gradient Boosting, while Logistic Regression remained bias-limited. Validation curves confirmed that the selected hyperparameters lie in stable operating regions.

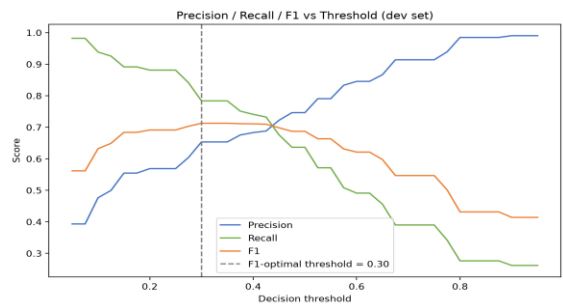


4. Calibration & Threshold Selection

Probability calibration was evaluated using calibration curves and Brier Score. Isotonic calibration achieved the best calibration performance. Although the F1-optimal threshold was approximately 0.30, the deployment threshold was set to 0.57 to support the project's precision-first objective.

Calibration Method	Brier Score
Uncalibrated	0.0887
Sigmoid	0.0887
Isotonic	0.0872 (Selected)

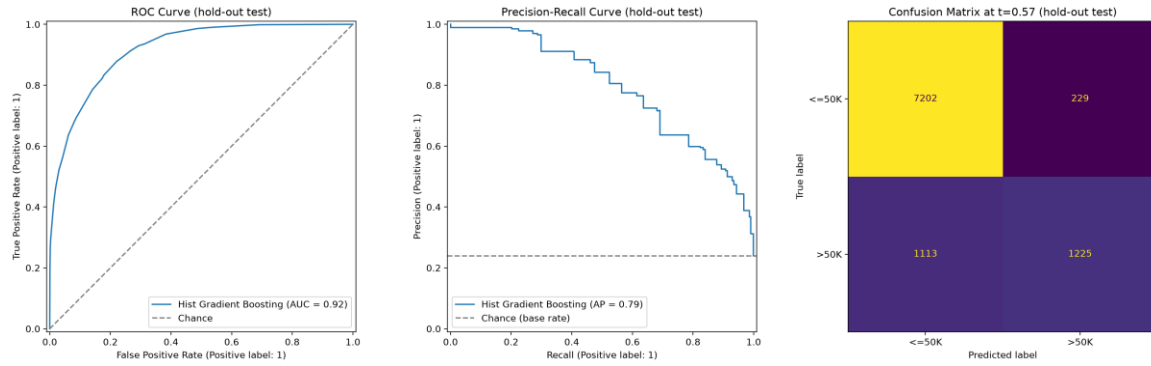
Threshold	Precision	Recall	Purpose
0.30	0.653	0.784	F1-optimal operating point
0.57	0.843	0.524	Selected deployment threshold
0.70	0.914	0.390	Very high precision example



5. Final Test Performance

Metric	Value
Accuracy	0.8626
Precision	0.8425
Recall	0.5240
F1-score	0.6461
ROC AUC	0.9173
PR AUC	0.7916

The tuned Histogram Gradient Boosting model maintained strong ranking performance while achieving high deployment precision. Lower recall is an intentional trade-off caused by the higher deployment threshold.



6. Saved Artifact

The saved Joblib artifact bundles feature engineering, preprocessing, the tuned Histogram Gradient Boosting model, isotonic calibration, deployment threshold, hyperparameters and metadata required for reproducible inference.

7. Conclusion

Histogram Gradient Boosting remained the strongest model after tuning and calibration. The final pipeline aligns with the project's precision-first objective while remaining fully reproducible. Future production deployments should monitor data drift and perform fairness audits across demographic groups.